课件网址：GitHub.com/styluck/RA
第一次作业：2024年11月4日 上交

$\mathbb{R}$ is both close and open. Its complement set ϕ is also both close and open.

Defn: (Open set) A set U is an open set, the for any $x \in U$, there exists $\varepsilon > 0$, such that

$$(x-\varepsilon, x+\varepsilon) \subseteq U$$

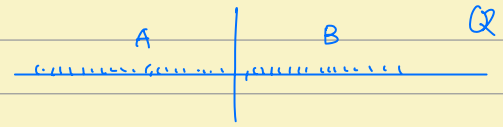
若P成立，则Q成立，当P为假时，这个蕴含式恒为真。

$0.9999\ldots = 1$

$\mathbb{R}$: the set of real numbers

$\mathbb{Q}$: the set of rational numbers

Dedekind cut 戴德金分割



① $A \cap B = \phi$

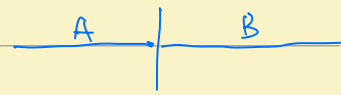
② $A \cup B = \mathbb{Q}$

③ $a \in A$ $b \in B$, then $a < b$

There exist three different cases:

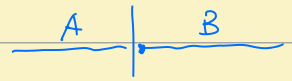
Case 1: $\sup A \in A$, and $\inf B \notin B$ A中存最大元素，B中无最小元素

$A = \{x \mid x \in \mathbb{Q}, x \leq 2\}$ $B = \{x \mid x \in \mathbb{Q}, x > 2\}$



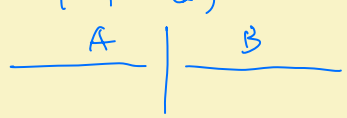
Case 2: $\sup A \notin A$, and $\inf B \in B$ A中无最大，B中有最小

$A = \{x \mid x \in \mathbb{Q}, x < 2\}$ $B = \{x \mid x \in \mathbb{Q}, x \geq 2\}$



Case 3: $\sup A \notin A$, and $\inf B \notin B$

$A = \{x \mid x \in \mathbb{Q}, x < 0 \text{ or } x^2 < 2\}$ $B = \{x \mid x \in \mathbb{Q}, x > 0 \text{ and } x^2 > 2\}$



~~Case~~ 4: $\sup A \in A$, $\inf B \in B$ (No possible)

$A \quad a \;\vdots\; b \quad B$

$a \neq b \quad a, b \in \mathbb{Q} \Rightarrow \frac{a+b}{2} \in \mathbb{Q}$ contradiction with $A \cup B = \mathbb{Q}$

Consider a cut $\mathbb{Q}_1 \Rightarrow A \mid B \to 0.999\ldots$

$\mathbb{Q}_2 \Rightarrow C \mid B \to 1$

$A = \{x \mid x \in \mathbb{Q},\ x < 0.999\ldots\}$

$C = \{x \mid x \in \mathbb{Q},\ x < 1\}$

We only need to prove $A = C$ $\Leftrightarrow$ $\begin{cases} \forall t \in A \Rightarrow t \in C \\ \forall t \in C \Rightarrow t \in A \end{cases}$

① Take $t \in A$ $\quad t < 0.999\ldots \Rightarrow t < 1 \Rightarrow t \in C$ $\quad \Rightarrow$

② Take $t \in C$ $\quad t < 1$ $\quad t = \frac{p}{q} < 1$, p, q are integer, $p < q$

$1 - t = 1 - \frac{p}{q} = \frac{q-p}{q} \geq \frac{1}{q}$

there exists $n > N$ such that $10^n > q \Rightarrow \frac{1}{10^n} < \frac{1}{q}$

$1 - t = \frac{q-p}{q} \geq \frac{1}{q} > \frac{1}{10^n} \Rightarrow 1 - t > \frac{1}{10^n}$

$t < 1 - \frac{1}{10^n} \Rightarrow t < 0.999\ldots \quad t \in A$ $\quad \Leftarrow$

2. Limits of functions

The limit of function f at x_0

$$\lim_{x \to x_0} f(x)$$

$\lim_{n \to \infty} f(n) \quad n \in \mathbb{N}$

The limit of function f as x goes to $+\infty$ or $-\infty$

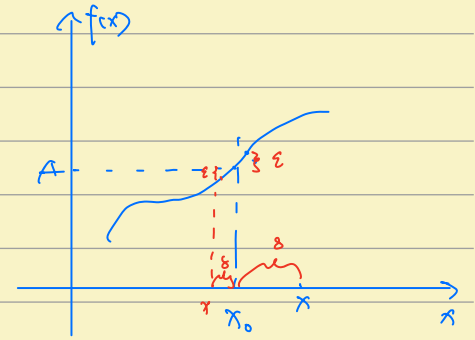
$$\lim_{x \to +\infty} f(x) \quad \text{or} \quad \lim_{x \to -\infty} f(x)$$

Definition 2.1 in Note

Defn (ε-δ-definition) Let $f: \mathbb{R} \to \mathbb{R}$ is a real-valued function. Suppose that $x_0 \in (b, c)$, and f is defined on (b, c) except possibly at x_0. The limit of f at x_0 is A if and only if for any $\varepsilon > 0$, there exists a $\delta > 0$, such that

$$|f(x) - A| < \varepsilon,$$

whenever $0 < |x - x_0| < \delta$.



Example: The derivative

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

The derivative of $f(x) = x^2$ $\quad x = x_0 + h$

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

$$\lim_{h \to 0} \frac{\cancel{x_0^2} + 2x_0 h + h^2 - \cancel{x_0^2}}{h} = \lim_{h \to 0} \frac{2x_0 h + h^2}{h}$$

$$= \lim_{h \to 0} 2x_0 + h = 2x_0$$

Example 2. Show that $\lim_{x\to 8} f(x) = x = 8$

Proof: For any $\varepsilon > 0$, there exists a $\delta > 0$ such that $|x-8| < \delta$ $\Rightarrow$ $|x-8| < \varepsilon$. Then we can take $\delta = \varepsilon$.

$|x - x_0| < \delta \Rightarrow$
$|f(x) - A| < \varepsilon$

Example 3 Show that $\lim_{x\to 8} (10x-2) = 78$

Proof: For any $\varepsilon > 0$, there exists a $\delta > 0$, such that $|x-8| < \delta \Rightarrow$
$|(10x-2) - 78| = |10x - 80| < \varepsilon \Rightarrow |x-8| < \frac{\varepsilon}{10}$
Then we can take $\delta = \frac{\varepsilon}{10}$. Then by definition, $\lim_{x\to 8}(10x-2) = 78$.

Example 4. $f(x) = \frac{x^2-4}{x-2}$ Show that $\lim_{x\to 2} f(x) = 4$

Proof: $|f(x) - 4| = \left|\frac{x^2-4}{x-2} - 4\right|$. Suppose $x \neq 2$, then we have

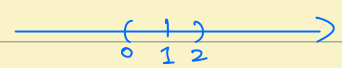
$$\left|\frac{x^2-4}{x-2} - 4\right| = |x+2-4| = |x-2| < \varepsilon.$$

Then we can take $\delta = \varepsilon$. Then by definition, $\lim_{x\to 2} f(x) = 4$

Example 5. Show that $\lim_{x\to 1} \frac{x^2-1}{2x^2-x-1} = \frac{2}{3}$.

Suppose $x \neq 1$, then $\left|\frac{x^2-1}{2x^2-x-1} - \frac{2}{3}\right| = \left|\frac{x+1}{2x+1} - \frac{2}{3}\right| = \frac{|x-1|}{3|2x+1|} \underline{\underline{<\varepsilon}}$

Let $0 < |x-1| < 1$ $\quad |x-1| < 1$



$\Rightarrow x > 0 \Rightarrow |2x+1| > 1$

Then we can take $\delta = \min\{3\varepsilon, 1\}$, then we have

$\delta = \min\{\frac{3}{2}\varepsilon, \frac{1}{2}\}$

$\frac{|x-1|}{3|2x+1|} < \frac{|x-1|}{3} < \varepsilon$, by definition, $\lim_{x\to 1} \frac{x^2-1}{2x^2-x-1} = \frac{2}{3}$

$\forall \varepsilon > 0 \quad \exists \delta$ s.t. $|x - x_0| < \delta \Rightarrow |f(x) - A| < \varepsilon$

① δ 依赖于 ε，ε 越小，则 δ 越小，但，δ 取更小也无妨。

② 只需求 $f(x)$ 在 $(x_0-\delta, x_0)\cup(x_0, x_0+\delta)$ 有定义即可，不需求在 $x=x_0$ 处有定义。

$\forall\, \varepsilon>0,\ \exists\, \delta>0$ s.t. $|x-x_0|<\delta \Rightarrow |f(x)-A|<\varepsilon$

Example 6. Show that $\lim\limits_{x\to 8} x^2 = 64$

Answer 1: $|x^2-64| = |x-8|\,|x+8| < \varepsilon$ Let $|x-8|<1 \Rightarrow |x|<9$

$\Rightarrow |x+8|<17$

then we have $|x-8|\,|x+8| < |x-8|\cdot 17 < \varepsilon$, take $\delta=\min(1, \frac{\varepsilon}{17})$

Then by definition, we have $\lim\limits_{x\to 8} x^2 = 64$

$|x-8|<\delta \Rightarrow -\delta < x-8 < \delta$

$\Rightarrow |x| < \delta+8 \Rightarrow |x+8| < \delta+16$

Answer 2: Suppose $|x-8|<\delta$, $|x+8|\,|x-8| < |\delta+16|\cdot\delta < \varepsilon$

$(\delta+16)\delta < \varepsilon \Rightarrow$ we take $\delta < -8+\sqrt{64+\varepsilon}$

Then by definition, we have $\lim\limits_{x\to 8} x^2 = 64$

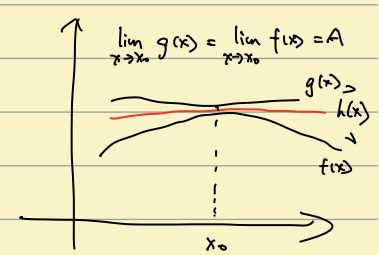
Thm 2.6 (Squeeze Theorem or Sandwich Theorem). Suppose $\alpha>0$, for all $x\in(x_0-\alpha, x_0)\cup(x_0, x_0+\alpha)$, such that

$$f(x)\le h(x)\le g(x)$$

and

$$\lim_{x\to x_0} f(x) = \lim_{x\to x_0} g(x) = A$$

then $\lim\limits_{x\to x_0} h(x) = A$.



Proof: For any $\varepsilon>0$, by definition, there exists $\delta_1>0$, such that

$|x-x_0|<\delta_1 \Rightarrow |f(x)-A|<\varepsilon \Leftrightarrow A-\varepsilon<f(x)<A+\varepsilon$

For any $\varepsilon>0$, by definition, there exists $\delta_2>0$, such that

$|x-x_0|<\delta_2 \Rightarrow |g(x)-A|<\varepsilon \Leftrightarrow A-\varepsilon<g(x)<A+\varepsilon$

Proof: For any $\varepsilon > 0$, by definition, there exists $\delta_1 > 0$, such that

$$|x - x_0| < \delta_1 \Rightarrow |f(x) - A| < \varepsilon \Leftrightarrow A - \varepsilon < f(x) < A + \varepsilon$$

For any $\varepsilon > 0$, by definition, there exists $\delta_2 > 0$, such that

$$|x - x_0| < \delta_2 \Rightarrow |g(x) - A| < \varepsilon \Leftrightarrow A - \varepsilon < g(x) < A + \varepsilon$$

Take $\delta = \min(\delta_1, \delta_2)$, we have

$$|x - x_0| < \delta \Rightarrow \quad A - \varepsilon < f(x) < A + \varepsilon$$

$$A - \varepsilon < g(x) < A + \varepsilon$$

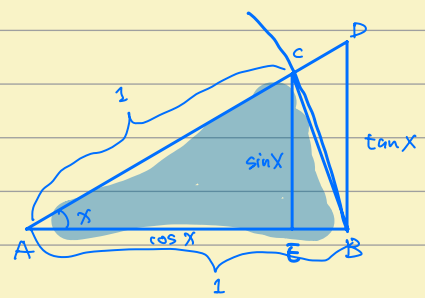
Since we have $f(x) \le h(x) \le g(x)$

$$\Rightarrow \quad A - \varepsilon < \cancel{f(x)} \le h(x) \le \cancel{g(x)} < A + \varepsilon$$

$$\Rightarrow \quad A - \varepsilon < h(x) < A + \varepsilon \Leftrightarrow |h(x) - A| < \varepsilon$$

By definition, we conclude $\lim\limits_{x \to x_0} h(x) = A$

Thm 2.7 ① $\lim\limits_{x \to 0} \dfrac{\sin x}{x} = 1$ ✓ ② $\lim\limits_{x \to +\infty} \left(1 + \dfrac{1}{x}\right)^x = e$ 二个重要极限



Proof the area of $\triangle ABC = \dfrac{\sin x}{2}$

the area of $\triangle ABD = \dfrac{\tan x}{2}$

the area of segment $\overset{\frown}{ABC} = \dfrac{x}{2}$

$$\triangle ABC < \overset{\frown}{ABC} < \triangle ABC$$

$$\Rightarrow \frac{\sin x}{2} < \frac{x}{2} < \frac{\tan x}{2}$$

$$\Rightarrow \sin x < x < \tan x$$

$$\Rightarrow \begin{cases} 1 < \dfrac{x}{\sin x} < \dfrac{1}{\cos x} \Rightarrow \cos x < \dfrac{\sin x}{x} < 1 \\ \lim\limits_{x \to 0} \cos x = \lim\limits_{x \to 0} 1 = 1 \end{cases}$$

$$\frac{CE}{DB} = \frac{AE}{AB} \qquad z = \frac{\sin x}{\cos x}$$

$$\frac{\sin x}{z} = \frac{\cos x}{1} = \tan x$$

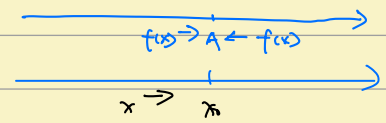
By the squeeze theorem, we have $\lim\limits_{x \to 0} \dfrac{\sin x}{x} = 1$

单侧极限

2.2. One-sided limits and limits as x tends to ∞

Defn 2.10 (One-sided limit) Suppose the domain of f contains $(x_0-\alpha, x_0)$ for some $\alpha>0$, and $A\in\mathbb{R}$. For all $\varepsilon>0$, there exists $\delta>0$, and $\delta<\alpha$, such that

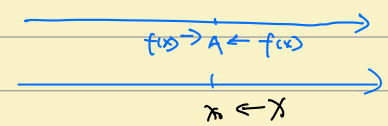
$$x_0-\delta<x<x_0 \Rightarrow |f(x)-A|<\varepsilon$$



We write $\lim\limits_{x\to x_0^-} f(x)=A$

Defn 2.10 (One-sided limit) Suppose the domain of f contains $(x_0, x_0+\alpha)$ for some $\alpha>0$, and $A\in\mathbb{R}$. For all $\varepsilon>0$, there exists $\delta>0$, and $\delta<\alpha$, such that

$$x_0<x<x_0+\alpha \Rightarrow |f(x)-A|<\varepsilon$$



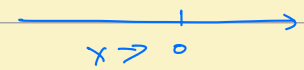
We write $\lim\limits_{x\to x_0^+} f(x)=A$

Prop 2.11. The limit $\lim\limits_{x\to x_0} f(x)=A$ exists, if and only if

① both $\lim\limits_{x\to x_0^+} f(x)$ and $\lim\limits_{x\to x_0^-} f(x)$ exist

② $\lim\limits_{x\to x_0^-} f(x) = \lim\limits_{x\to x_0^+} f(x) = A$

Example 1 Prove that $\lim\limits_{x\to 0^-} e^{\frac{1}{x}} = 0$



Proof: For any $\varepsilon>0$. $|e^{\frac{1}{x}}|<\varepsilon \Rightarrow |\frac{1}{x}|<|\ln\varepsilon|$

$$\Rightarrow -\frac{1}{|\ln\varepsilon|}<x<\frac{1}{|\ln\varepsilon|}$$

Since $x<0$, let $\delta=\frac{1}{-|\ln\varepsilon|}=-\frac{1}{\ln\varepsilon}>0$

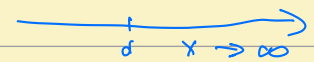
$$-\delta<x<0 \Rightarrow |e^{\frac{1}{x}}-0|<\varepsilon.$$

By definition, we obtain $\lim\limits_{x\to 0^-} e^{\frac{1}{x}}=0$

Defn 2.13 (x tends to ∞) Suppose $f: \mathbb{R} \to \mathbb{R}$ is real-valued function, the domain of f contains $(\alpha, +\infty)$, for some $\alpha \in \mathbb{R}$, $A \in \mathbb{R}$. If for any $\varepsilon > 0$, there exist $K > \alpha$, such that

$$\text{for any } x > K \Rightarrow |f(x) - A| < \varepsilon$$

We write $\lim_{x \to \infty} f(x) = A$



Similarly, we can define

$f(x) = \frac{1}{x}$

$f(x) \in (0, +\infty)$ when $x \in (0, +\infty)$

$$\lim_{x \to -\infty} f(x) = A$$

$$\lim_{|x| \to \infty} f(x) = A$$

$f(x) = \frac{1}{x^2}$